

(h) Estimation		
2h.01	<u>Use calculated or given summary statistical data to make estimates of population characteristics. Use samples to estimate population mean. Use sample data to predict population proportions.</u>	e.g. predict that approximately half the population will be above the sample median.
2h.02	(Higher tier topic only; not assessed at Foundation tier)	
2h.03	Know that sample size has an impact on reliability and replication.	e.g. know that results/conclusions are likely to be more reliable if based on larger samples.

What students need to learn:		Guidance
3. Probability		
3p.01	Use collected data to calculate estimates of probabilities.	Use of relative frequency.
3p.02	Compare the probability of different possible outcomes using the 0–1 or 0–100% scale and statements of likelihood.	Locate events on a probability scale and use the language of likelihood (e.g. certain, impossible, evens, likely, very unlikely, etc.)
3p.03	Use probability values to calculate expected frequency of a specified characteristic within a sample or population.	Given total frequency, use probability as a proportion to find expected frequency.
3p.04	<u>Use collected data and calculated probabilities to determine and interpret relative risks and absolute risks, and express in terms of expected frequencies in groups.</u>	e.g. use driving test pass rate data with Instructor A and Instructor B to determine the probability (absolute risk) of passing with A, or determine the relative probability (relative risk) of passing with A compared with B. $\text{Relative risk} = \frac{\text{risk of passing with A}}{\text{risk of passing with B}}$
3p.05	Compare experimental data with theoretical predictions to identify possible bias within the experimental design.	e.g. consider whether a set of dice rolls suggest that the dice is fair
3p.06	Recognise that experimental probability will tend towards theoretical probability as the number of trials increases when all variables are random.	Understand that increasing sample size generally leads to better estimates of probability and population parameters. Students may be expected to estimate probabilities from relative frequency diagrams and frequency tables.
3p.07	<u>Use two-way tables, sample space diagrams, tree diagrams and Venn diagrams to represent all the different outcomes possible for at most three events.</u>	<u>Use of these for conditional probability is expected.</u> (See 3p.09.) <u>Sample space diagrams may include listing or tabulating all outcomes of single events, or successive events, in a systematic way.</u> <u>Understand the terms mutually exclusive and exhaustive.</u> <u>Know the addition law for two mutually exclusive events:</u> $P(A \text{ or } B) = P(A) + P(B)$

Foundation tier content

What students need to learn:		Guidance
3. Probability <i>continued</i>		
3p.08	<u>Know and apply the formal notation for independent events.</u>	<u>Understand the difference between independent and conditional events.</u> <u>The multiplication law for independent events must be known:</u> $P(A \text{ and } B) = P(A) \times P(B)$ <u>Know that for independent events:</u> $P(A B) = P(A) \text{ and } P(B A) = P(B)$
3p.09	<u>Know and apply the formal notation for conditional probability.</u>	<u>The formula for conditional probability must be known:</u> $P(B A) = \frac{P(A \text{ and } B)}{P(A)}$
3p.10 to 3p.13	(Higher tier topics only; not assessed at Foundation tier)	

Higher tier content

A variety of contexts, including real-life data, will be used. (No detailed knowledge of those contexts will be expected.) Each item of content will have a code, for example **1a.01**, where **1** is 'The collection of data', **a** is 'Planning' and **01** is the numerical order of the content item.

What students need to learn:	Guidance
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